

① 143.75 (decimal) to Binary.

$$2 \overline{) 143} \quad 1$$

$$2 \overline{) 71} \quad 1$$

$$2 \overline{) 35} \quad 1$$

$$2 \overline{) 17} \quad 1$$

$$2 \overline{) 8} \quad 0$$

$$2 \overline{) 4} \quad 0$$

$$2 \overline{) 2} \quad 0$$

$$1$$

$$0.75 = 0.11$$

$$\text{Binary: } 10001111.11_2$$

143.75 (decimal) to Octal.

$$10001111.11_2$$

$$\underline{8 \overline{) 143}} \quad 010 \ 001 \ 111.110$$

$$217.68$$

143.75 (decimal) to hexadecimal.

$$1000 \ 1111.1100$$

$$= 8F.C_{16}$$

② Binary: 110111012 to decimal.

$$2^4 \times 1 + 2^3 \times 1 + 2^2 \times 0 + 2 \times 1 + 1 + \frac{1}{2} + \frac{0}{4} + \frac{1}{8}$$

$$= 27.625$$

②

Octal: 011 011 · 101 ·
33 · 5₈ ·

Hexadecimal: 0001 1011 · 1010 ·
1B · A₁₆ ·

③ Octal: 453₈ · to decimal ·

$$4 \times 8^2 + 5 \times 8 + 3 = 299 \cdot$$

Binary

- 4 → 100
 - 5 → 101
 - 3 → 011
- $$= 100101011_2$$

Hexadecimal:

Binary: 100101011 ·
0001 0010 1011 ·
= 12B₁₆ ·

④

Hex: 1F · C₁₆ ·

Decimal: $1 \times 16 + 15 = 31 \cdot$

$$\text{Fraction} = \frac{12}{16} = 0.75 \\ = 31.75_{10} \cdot$$

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Binary : $1 \rightarrow 0001$

$F \rightarrow 1111$

$C \rightarrow 1100$

$= 1111.11$

Decimal : $001.111.110$

$= 32.68$

Table :

	Decimal	Binary	Octal	Hexa
①	143.75	10001111.11	214.6	8F.C
②	27.625	11011.101	33.5	1B.A
③	299	100101011	453	12B
④	31.75	11111.11	37.6	1F.C

②

(a) $3245 = ?_{10}$

$$3245 = 5^2 \times 3 + 5^1 \times 2 + 5^0 \times 4$$

$$= 89_{10}$$

(b) $32.234 = ?_{10}$

$$32.234 = 4^1 \times 3 + 4^0 \times 2 + 2 \times 4^{-1} + 3 \times 4^{-2}$$

$$= 14.6875_{10}$$

④

$$(c) 2310 = 44$$

$$7 \overline{) 231} 0$$

$$7 \overline{) 33} 5$$

$$7 \overline{) 4} 4$$

0

$$= 4804$$

$$(d) 18910 = 96$$

$$6 \overline{) 189} 3$$

$$6 \overline{) 31} 1$$

$$6 \overline{) 35} 5$$

0

$$= 5136$$

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Q(3) Find the radix r .

$$(a) \frac{312r}{20r} = 13.1r.$$

$$312r = 3r^2 + r + 2.$$

$$20r = 2r.$$

$$13.1r = r + 3 + 1/r.$$

$$\frac{3r^2 + r + 2}{2r} = r + 3 + \frac{1}{r}.$$

$$r^2 - 3r = 0.$$

$$\boxed{r = 3}$$

$$(b) 264r + 136r = 433r.$$

$$(2r^2 + 6r + 4) + (r^2 + 3r + 6) = 4r^2 + 3r + 3.$$

$$r^2 - 6r - 7 = 0.$$

$$(r - 7)(r + 1) = 0.$$

$$r = 7.$$

$$(c) 2215058 = 12345r.$$

$$2215058 = 44565r$$

$$r^4 + 2r^3 + 3r^2 + 4r + 5 = 44565r$$

⑥

$$16^4 + 4(16^3) + 1(16^2) + 4(16) + 5 = 74565$$

$$\boxed{r=16}$$

④ (a) $1101_2 + 1101111_2$

$$\begin{array}{r} 00001101 \\ + 11011111 \\ \hline 11101100_2 \end{array} \Rightarrow \boxed{11101100_2}$$

(b) $77652_8 + 76543_8$

$$\begin{array}{r} 77652 \\ + 76543 \\ \hline 176415_8 \end{array} \Rightarrow \boxed{176415_8}$$

(c) $578A_{16} + ABDE_{16}$

$$\begin{array}{r} 578A \\ + ABDE \\ \hline 10368_{16} \end{array} \Rightarrow \boxed{10368_{16}}$$

(d) $234_6 + 315_6$

$$\begin{array}{r} 234 \\ + 315 \\ \hline 551_6 \end{array} \Rightarrow \boxed{551_6}$$

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⑤ Binary $\rightarrow$ Gray code.

(a) 10101_2 .

Binary	1	0	1	0	1
Gray	1	1	1	1	1

Ans :

11111

(b) 01110_2 .

Binary	0	1	1	1	0
Gray	0	1	0	0	1

Ans :

01001

⑥ Gray code $\rightarrow$ Binary.

(a) 01101

Gray	0	1	1	0	1
Binary	0	1	0	0	1

Ans :

01001

(b) 10101

Gray	1	0	1	0	1
Binary	1	1	0	0	1

Ans :

11001